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Java SE 27 & JDK 27

Class OutputStreamWriter

java.lang.Object
    java.io.Writer
        java.io.OutputStreamWriter

All Implemented Interfaces:

Closeable, Flushable, Appendable, AutoCloseable

Direct Known Subclasses:

FileWriter

```
public class OutputStreamWriter
extends Writer
```

An OutputStreamWriter is a bridge from character streams to byte streams: Characters written to it are encoded into bytes using a specified `charset`. The charset that it uses may be specified by name or may be given explicitly, or the default charset may be accepted.

Each invocation of a write() method causes the encoding converter to be invoked on the given character(s). The resulting bytes are accumulated in a buffer before being written to the underlying output stream. Note that the characters passed to the write() methods are not buffered.

For top efficiency, consider wrapping an OutputStreamWriter within a BufferedWriter so as to avoid frequent converter invocations. For example:

```
    Writer out = new BufferedWriter(new OutputStreamWriter(anOutputStream));
```

A *surrogate pair* is a character represented by a sequence of two `char` values: A *high* surrogate in the range '\uD800' to '\uDBFF' followed by a *low* surrogate in the range '\uDC00' to '\uDFFF'.

A *malformed surrogate element* is a high surrogate that is not followed by a low surrogate or a low surrogate that is not preceded by a high surrogate.

This class always replaces malformed surrogate elements and unmappable character sequences with the charset's default *substitution sequence*. The CharsetEncoder class should be used when more control over the encoding process is required.

Since:

1.1

See Also:

BufferedWriter, OutputStream, Charset

Field Summary

Fields declared in class Writer
lock

Constructor Summary

Constructors

Constructor	Description
OutputStreamWriter (OutputStream out)	Creates an OutputStreamWriter that uses the default character encoding, or where out is a PrintStream, the charset used by the print stream.
OutputStreamWriter (OutputStream out, String charsetName)	Creates an OutputStreamWriter that uses the named charset.
OutputStreamWriter (OutputStream out, Charset cs)	Creates an OutputStreamWriter that uses the given charset.
OutputStreamWriter (OutputStream out, CharsetEncoder enc)	Creates an OutputStreamWriter that uses the given charset encoder.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
void	close()	Closes the stream, flushing it first.
void	flush()	Flushes the stream.
String	getEncoding()	Returns the name of the character encoding being used by this stream.

void	write(char[] cbuf, int off, int len)	Writes a portion of an array of characters.
void	write(int c)	Writes a single character.
void	write(String str, int off, int len)	Writes a portion of a string.

Methods declared in class Writer

append, append, append, nullWriter, write, write

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out,
                          String charsetName)
                   throws UnsupportedEncodingException
```

Creates an OutputStreamWriter that uses the named charset.

Parameters:

out - An OutputStream

charsetName - The name of a supported charset

Throws:

UnsupportedEncodingException - If the named encoding is not supported

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out)
```

Creates an OutputStreamWriter that uses the default character encoding, or where out is a PrintStream, the charset used by the print stream.

Parameters:

out - An OutputStream

See Also:

Charset.defaultCharset()

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out,
                          Charset cs)
```

Creates an OutputStreamWriter that uses the given charset.

Parameters:

`out` - An OutputStream

`cs` - A charset

Since:

1.4

OutputStreamWriter

```
public OutputStreamWriter(OutputStream out,
                          CharsetEncoder enc)
```

Creates an OutputStreamWriter that uses the given charset encoder.

Parameters:

`out` - An OutputStream

`enc` - A charset encoder

Since:

1.4

Method Details

getEncoding

```
public String getEncoding()
```

Returns the name of the character encoding being used by this stream.

If the encoding has an historical name then that name is returned; otherwise the encoding's canonical name is returned.

If this instance was created with the `OutputStreamWriter(OutputStream, String)` constructor then the returned name, being unique for the encoding, may differ from the name passed to the constructor. This method may return `null` if the stream has been closed.

Returns:

The historical name of this encoding, or possibly `null` if the stream has been closed

See Also:

`Charset`

write

```
public void write(int c)
           throws IOException
```

Writes a single character.

Overrides:

`write` in class `Writer`

Parameters:

`c` - int specifying a character to be written

Throws:

`IOException` - If an I/O error occurs

write

```
public void write(char[] cbuf,
                  int off,
                  int len)
           throws IOException
```

Writes a portion of an array of characters.

Specified by:

`write` in class `Writer`

Parameters:

`cbuf` - Buffer of characters

`off` - Offset from which to start writing characters

`len` - Number of characters to write

Throws:

`IndexOutOfBoundsException` - If `off` is negative, or `len` is negative, or `off + len` is negative or greater than the length of the given array

`IOException` - If an I/O error occurs

write

```
public void write(String str,
                  int off,
                  int len)
           throws IOException
```

Writes a portion of a string.

Overrides:

`write` in class `Writer`

Parameters:

`str` - A String

`off` - Offset from which to start writing characters

`len` - Number of characters to write

Throws:

`IndexOutOfBoundsException` - If `off` is negative, or `len` is negative, or `off + len` is negative or greater than the length of the given string

`IOException` - If an I/O error occurs

flush

```
public void flush()
           throws IOException
```

Flushes the stream.

Specified by:

`flush` in interface `Flushable`

Specified by:

`flush` in class `Writer`

Throws:

`IOException` - If an I/O error occurs

close

```
public void close()
           throws IOException
```

Description copied from class: `Writer`

Closes the stream, flushing it first. Once the stream has been closed, further write() or flush() invocations will cause an IOException to be thrown. Closing a previously closed stream has no effect.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Specified by:

`close` in class `Writer`

Throws:

`IOException` - If an I/O error occurs

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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